Appendices D & E: The Crossover Point and a Zero-Free-Parameter Law $s^*(\gamma)$

UIP Phase 1 Review Packet · Confirmed preregistered predictions · Companion to preprint DOI: 10.5281/zenodo.21246247 · July 2026
Code: **appendix_d_crossover.py**, **appendix_e_gamma_sweep.py** at github.com/derekhone/uip-phase1-testbeds

Headline result. The framework's benefit/cost bookkeeping implies a critical coupling strength s* below which correcting the accessible relay qubit costs more transferable coherence than it delivers. Appendix D located this crossover in advance at a single operating point. Appendix E then derived the full functional law with **zero free parameters** — $s^*(\gamma) = \arcsin(\sqrt{\gamma(1-\bar{C}^*)/\bar{C}^*})$ with $\bar{C}^* = 1/1.45$ — and confirmed it across a sweep of dephasing rates with **r = 0.976** between predicted and measured crossover points, each inside its preregistered ±0.05 window. The restricted-access advantage also generalized to a three-qubit chain. These are the strongest confirmed predictions in the Phase 1 record. Their scope is bounded by Appendices F and G: they are quantum-feedback-specific boundary predictions, not optimal-control or universal claims.

1. Setting

The restricted-access testbed of Appendix C: qubit A (accessible relay) coupled by partial SWAP of strength s (per-step transfer fraction $\tau = \sin^2(s)$) to qubit B (shielded target holding coherence). Dephasing attack alternates between A and B (γ_{hi}, $\gamma_{lo} = 0.01$, period 15, T = 90 cycles). The only available action is partial re-preparation of A toward the reference state (f = 0.5). Policies act on noisy estimates ($\sigma = 0.02$ on coherence, $\sigma = 0.01$ on rates); 30 seeds per condition.

2. Appendix D: the crossover point (preregistered, single operating point)

Reasoning: re-preparing A destroys part of the off-diagonal phase structure A is carrying toward B (measurement back-action). At weak coupling, transfer is slow, so a corrected A loses its refreshed coherence to noise before delivering it — correction is net-negative. At strong coupling, refreshed coherence reaches B before decaying — correction is net-positive. Hence a crossover s* must exist where the inheritance-guided policy's advantage over never-correct changes sign.

Preregistered prediction: at $\gamma_{hi} = 0.15$, the sign change occurs at $s^* = 0.26 \pm 0.05$, computed in advance from the benefit/cost balance G = L: delivered benefit $(1-\bar{C})\cdot\tau\cdot e^{-\gamma}$ equals disturbance cost $0.5\cdot R\cdot\bar{C}\cdot\tau$ at steady relay coherence $\bar{C}^* = 1/1.45 \approx 0.6897$, giving $s^* = \arcsin(\sqrt{\gamma(1-\bar{C}^*)/\bar{C}^*}) = 0.2628$.

Result: measured crossover s* = 0.271 (advantage interpolated to zero between sweep points). |error| = 0.008, inside the ±0.05 window. **PASS**.

3. Appendix E: the zero-free-parameter law $s^*(\gamma)$

Preregistered predictions (locked before running; the constant $\bar{C}^* = 1/1.45$ frozen from Appendix D, nothing re-tuned):

- **P-E1:** for each $\gamma \in \{0.08, 0.12, 0.15, 0.20, 0.25, 0.30\}$, the measured crossover lands within ± 0.05 of $s^*(\gamma)$ from the formula.
- **P-E2 (generalization):** in a three-qubit chain (accessible A → relay M → shielded target B), the inheritance-guided policy retains its advantage over all baselines above crossover.
- **Falsification:** any point outside its window, or loss of advantage in the chain → the law is an artifact of one operating point.

Results

γ_{hi}	Predicted $s^*(\gamma)$	Measured s^*	|error|	Verdict
0.08	0.1907	0.183	0.008	**PASS**
0.12	0.2343	0.241	0.007	**PASS**
0.15	0.2628	0.271	0.008	**PASS**
0.20	0.3054	0.318	0.013	**PASS**
0.25	0.3420	0.361	0.019	**PASS**
0.30	0.3775	0.402	0.025	**PASS**

P-E1: PASS — all six points inside their windows; correlation between predicted and measured crossovers r = 0.976. A mild systematic underprediction grows with γ (the formula's first-order decay factor $e^{-\gamma}$ under-counts higher-order losses), noted honestly; all points remain within window.

P-E2: PASS — three-qubit chain. Above crossover, the inheritance-guided policy preserved more target coherence than always-correct, never, and random baselines; below crossover, never-correct won, matching the boundary structure.

$$s^*(\gamma) = \arcsin\sqrt{\gamma(1-\bar{C}^*)/\bar{C}^*}, \quad \bar{C}^* = 1/1.45$$

This is the strongest object in the Phase 1 record: a functional relationship between an environmental parameter (γ) and a control boundary (s^*), containing **no fitted parameters**, confirmed across the swept range.

4. Scope, bounded by later appendices

- These are *boundary* predictions — where regimes change — not claims of optimal control. Appendix F subsequently showed the framework's quantities fall 12–22% short of a near-optimal adversary; that result stands alongside, not against, these.

- The constant $\bar{C}^* = 1/1.45$ encodes a measurement-back-action balance and does **not** transfer to classical systems (Appendix G); the law is quantum-feedback-specific.
- All results are consistent with standard quantum mechanics and feedback control. No new physics is claimed. Ladder placement: Level 4.

5. Reproduction and invitation

Code: `appendix_d_crossover.py`, `appendix_e_gamma_sweep.py` (numpy, scipy; deterministic up to seeded RNGs). Reviewers are invited to break the law outside the swept range — higher γ, non-Markovian noise, amplitude-damping-dominant channels — or on hardware using measured device rates. A refutation outside the tested range would be a welcome and publishable tightening of the record.

"A false balance is an abomination to the LORD, but a just weight is His delight." — Proverbs 11:1